Venkata Vinesh Kumar Reddy Atluri

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Available: compulsory 6-month ML / AI engineering internship, 1 March – 31 August 2027 (30 ECTS, ECL agreement)

Education

École Centrale de Lyon — Écully, France
Lyon Centrale Digital Lab 2026–2027, Diplôme d'établissement (final-year exchange) — *Sep 2026 – Aug 2027*

Mahindra University — Hyderabad, India
B.Tech in Computer Science and Engineering: 3 years completed, final year on exchange at ECL — *Aug 2023 – 2027 (expected)*

Technical Skills

Languages: Python (primary), TypeScript
ML and DL: PyTorch (1-D U-Net), time-series forecasting, GARCH volatility modelling, walk-forward evaluation, signal processing, model selection (BIC)
LLM and data: LLM prompting pipelines, retrieval over documents, NumPy/SciPy (least-squares optimisation)
Engineering: FastAPI, React, Next.js, Supabase, Docker, Qt, multiprocessing, pytest, Git

Projects

TriageMate, team hackathon project, Swiss {ai} Weeks 2026 | *Python, FastAPI, React, Docker, Azure* — Sep 2026

- Triage co-pilot for service desks (Swiss Life challenge): screens tickets for PII and prompt injection, classifies and routes them, assigns a justified priority, retrieves similar cases and drafts a reply an analyst approves. I built the classification and retrieval logic and the LLM prompting pipeline.
- Found that the 20,000-ticket set was only 173 unique descriptions with effectively random priority labels (a trained classifier matched always guessing the majority class); computed priority with an explicit ITIL urgency x impact matrix and used the tickets as a retrieval library, with the LLM only where judgment was needed.

PRECITEC Sensor Peak Detection, École Centrale de Lyon | *Code private (lab project); details on request* — Oct 2026

- Team of 4, for the LTDS lab. Designed and implemented a model-based peak-detection algorithm for a PRECITEC CHRocodile CLS 2 confocal line sensor (1,200 channels): measured-template fitting with quasi-BIC model selection, integrated live into the lab's Python app. I planned the project and built the detector, simulator and benchmark; teammates built acquisition and the viewer.
- Lab tests on the real sensor showed my first Gaussian detectors found 4–11 surfaces where there were 2, because real peaks are not Gaussian. The final detector finds exactly 2 surfaces on 120/120 real glass-slide channels (baseline Gaussian method: 4) and resolved a pair 0.77 peak widths apart on a live frame.
- On a fixed simulated benchmark: 93% of close pairs found at 0.75 peak widths, 97% of single surfaces and 100% of triples counted correctly. Derived the theoretical resolution limit (Cramér–Rao) and showed the detector reaches the limit set by current peak-shape knowledge.
- Retrained a 1-D U-Net on the realistic simulator (F1 0.57 to 0.927, 0.85 s per line); it invents surfaces on single peaks, so the robust detector makes the final call.

Maintenance Copilot (in progress) | *Python, PyTorch, FastAPI, Docker* — Nov 2026 – in progress

- Building an agent for industrial compressor maintenance that combines anomaly detection on real compressor telemetry (MetroPT-3), retrieval over public equipment manuals in English, French and German, and cited diagnoses with structured work orders, abstaining when the evidence is thin.
- Being evaluated on a strict time split, with synthetic fault scenarios always labelled synthetic. No results yet; they will be published with the public repository.

Veltrix, market analytics terminal | *TypeScript, Next.js, Supabase* | repo | live demo — Dec 2025 – Present

- Live terminal with a technical signal engine that exposes every indicator's vote, walk-forward-validated forecasts with calibrated bands, and paper trading on row-level-secured portfolios.
- Reported a 50.2% directional hit-rate over 1,500 out-of-sample walk-forward calls ($z = 0.16$) as no edge, rather than a tuned backtest. Replacing EWMA with GARCH(1,1) volatility raised 95% interval coverage from 91.67% to 94.75%.
- Built and benchmarked Ridge regression and gradient-boosted stumps, then cut them: RMSE changed by under 1% at 123–443 times slower inference.

Coursework, Certification and Languages

Relevant coursework: Deep Neural Networks, Reinforcement Learning, Computational Finance, Applied Time Series Analysis
Certification: Machine Learning Specialization, DeepLearning.AI and Stanford Online (Andrew Ng), Aug 2026. Credential ID JOWXFIWH9QN2.
Languages: English (fluent), Telugu (native), Hindi (conversational), French (A1, in progress)

Leadership

Mathematics Club, Mahindra University: Logistics Head (Aug 2024 – Apr 2025), then Head (Jul 2025 – May 2026). Ran logistics for university mathematics modelling competitions and academic events.